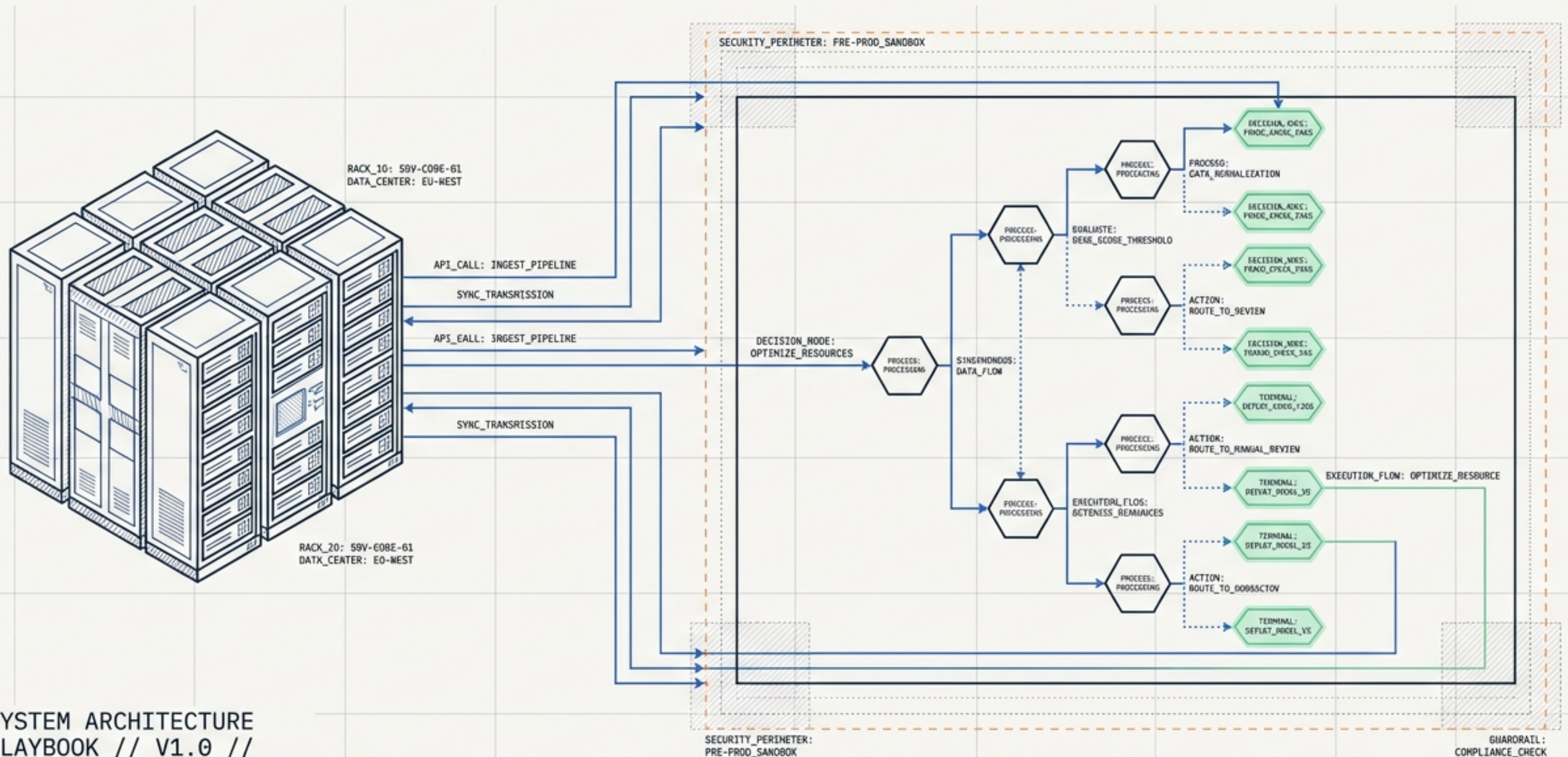


AI-Native Enterprise Engineering

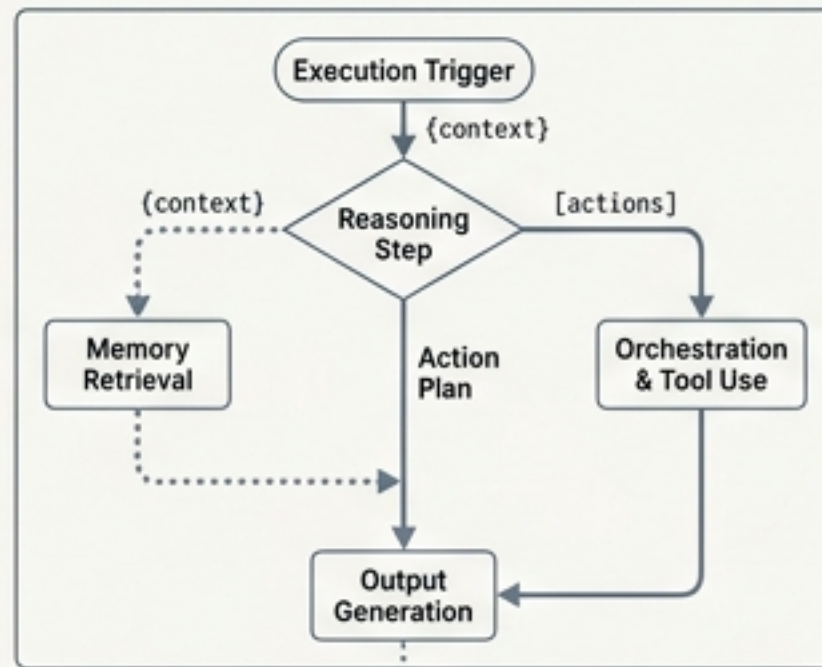


```
SYSTEM ARCHITECTURE
PLAYBOOK // V1.0 //
NO AUTONOMOUS PRODUCTION CHANGES
```


The LLM is just the brain; it requires a nervous system

Core Logic

Agent execution loops provide memory and orchestration.



The Brain:
LLM
(Intelligence & Reasoning)

Connective Tissue

APIs, gRPC, and MCP data access provide enterprise integration.

API Calls (REST/GraphQL)

/api/v1/query

/api/v1/query

gRPC Streams

service.GetSchema()

gRPC Streams

service.GetSchema()

MCP Data Access

service.GetSchema()

Enterprise Services

Data Warehouse

Microservices Mesh

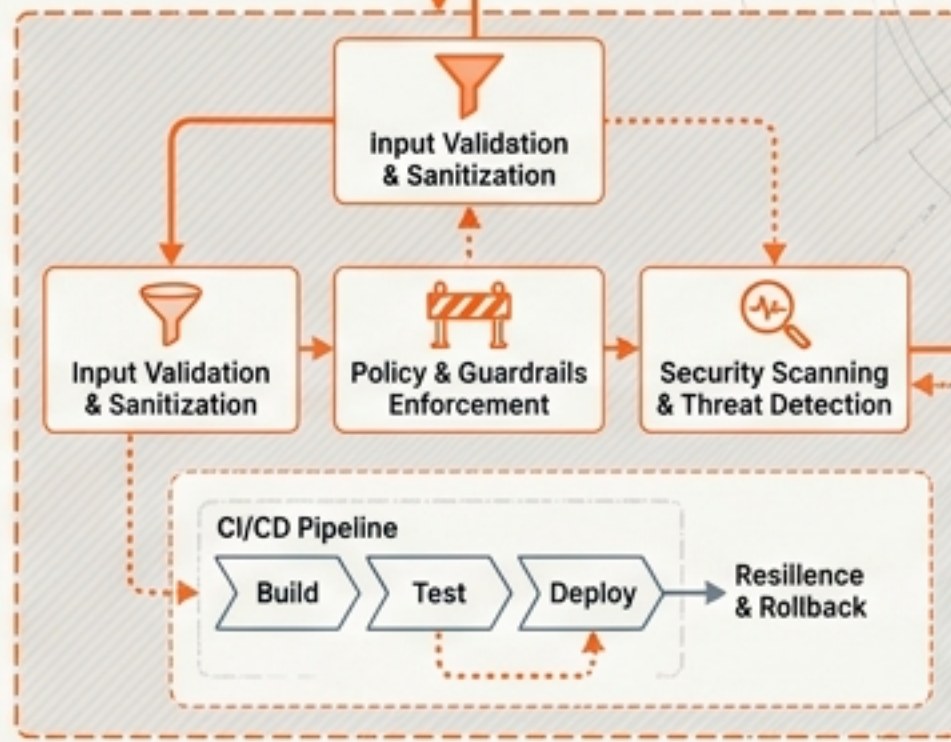
Enterprise Services

Data Warehouse

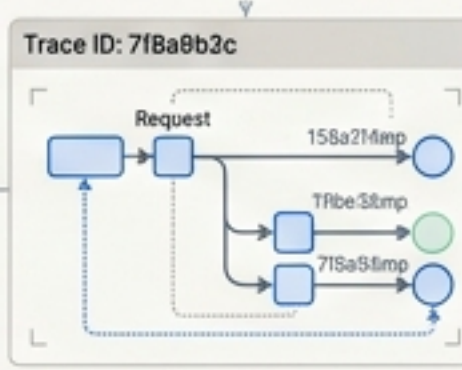
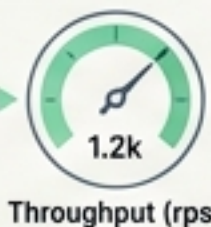
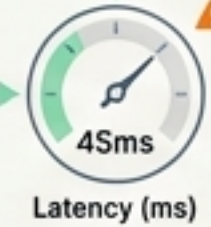
Microservices Mesh

Immune System

Security, guardrails, and CI/CD pipelines provide validation and resilience.



Human Oversight & Feedback Loop



Sensory Organs

Observability, tracing, and metrics provide monitoring and human oversight.

The continuous autonomous execution cycle

1. Perception

Gathers context.

- Sliding Windows
- Stream Processing
- PII filtering

2. Reasoning

Decides next steps.

- Prompt-as-Code
- Zod JSON validation
- Tool Selection

3. Action

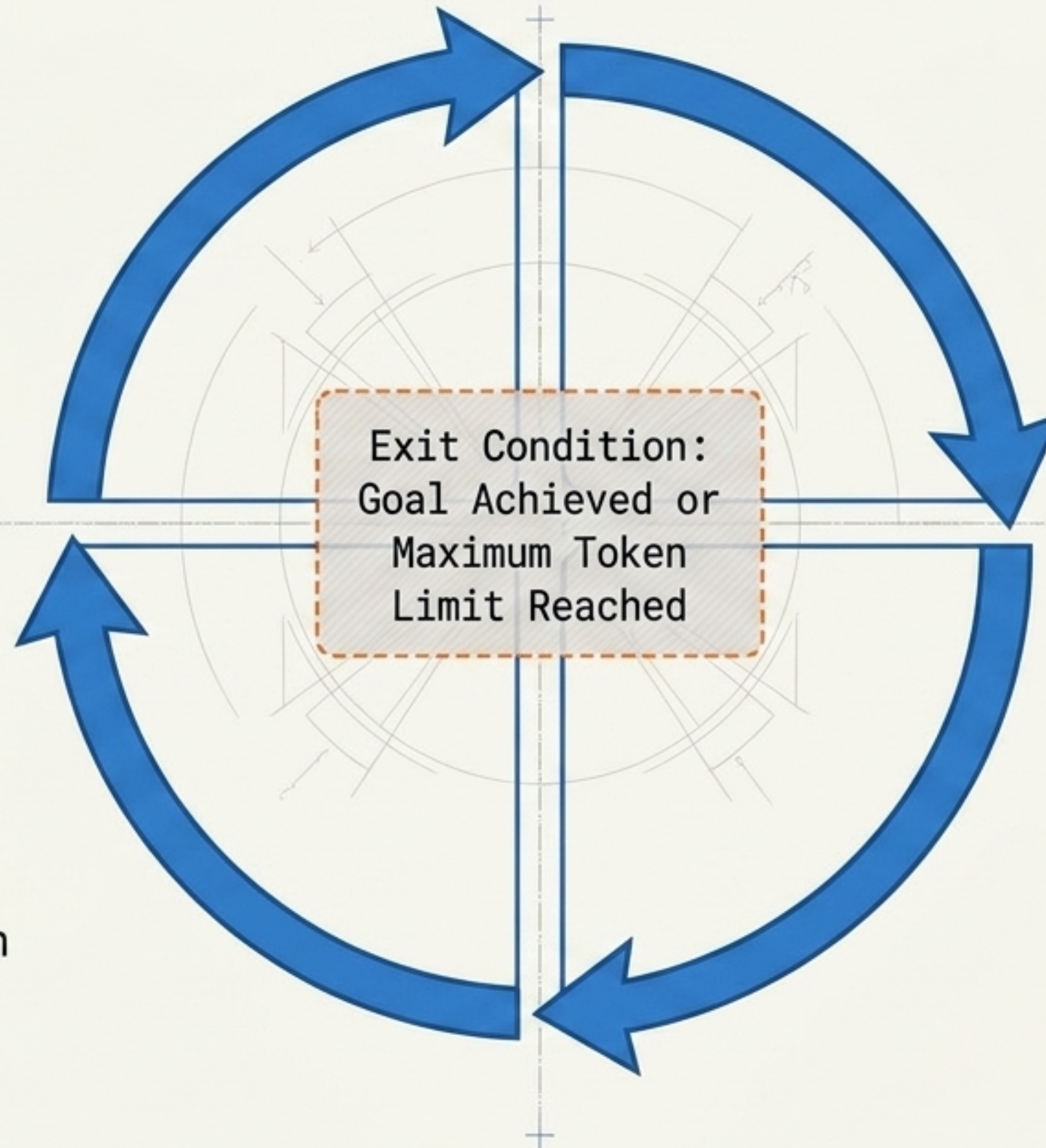
Executes operations.

- Idempotency keys
- Sandboxing
- Saga Pattern

4. Reflection

Evaluates outcomes.

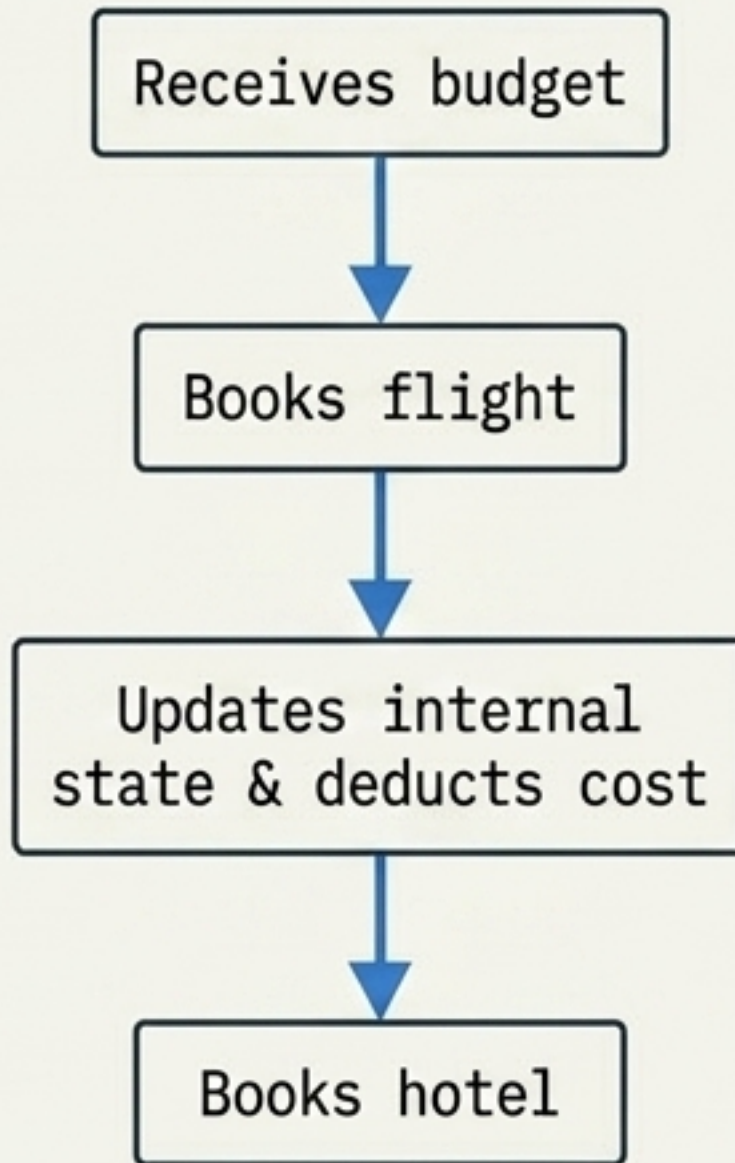
- Critic Models
- Event Sourcing
- Graceful Degradation



Domain-specific adaptations of the core agent loop

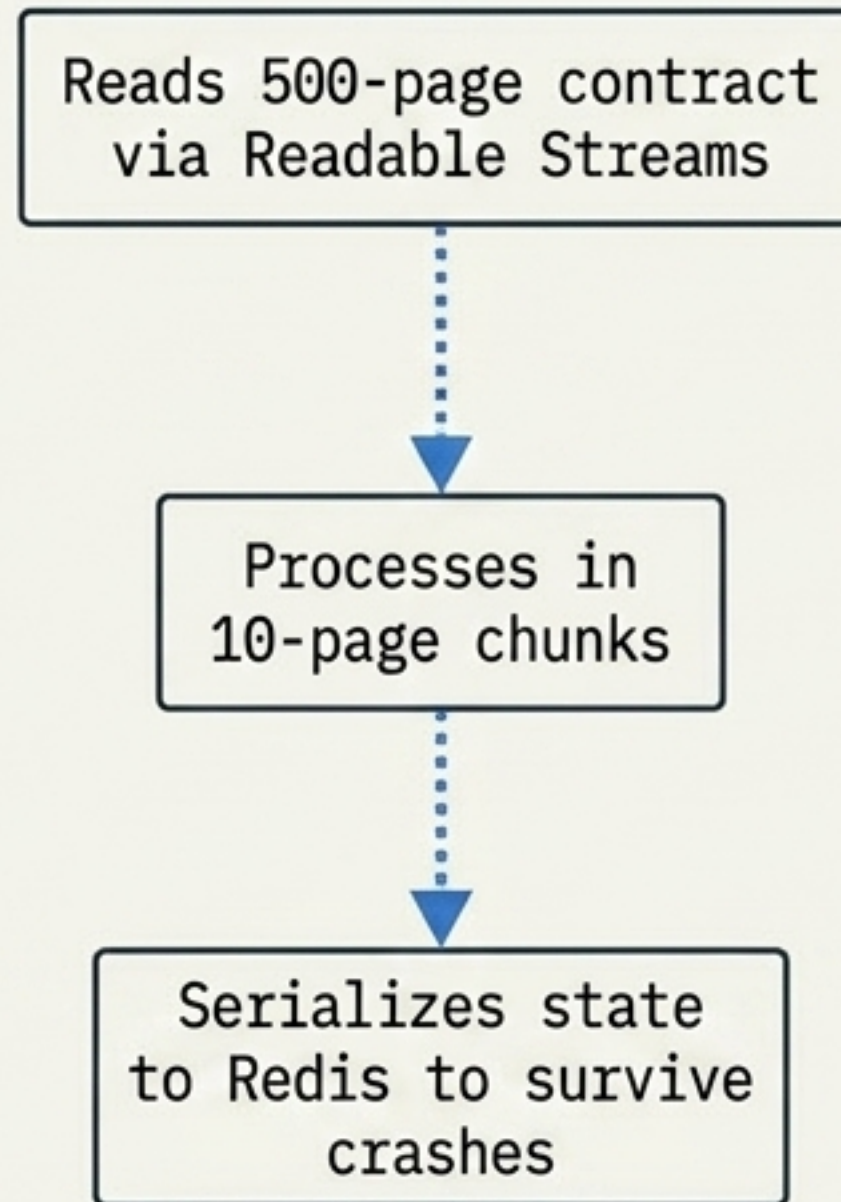
AI Travel Agent

Focus: State Management



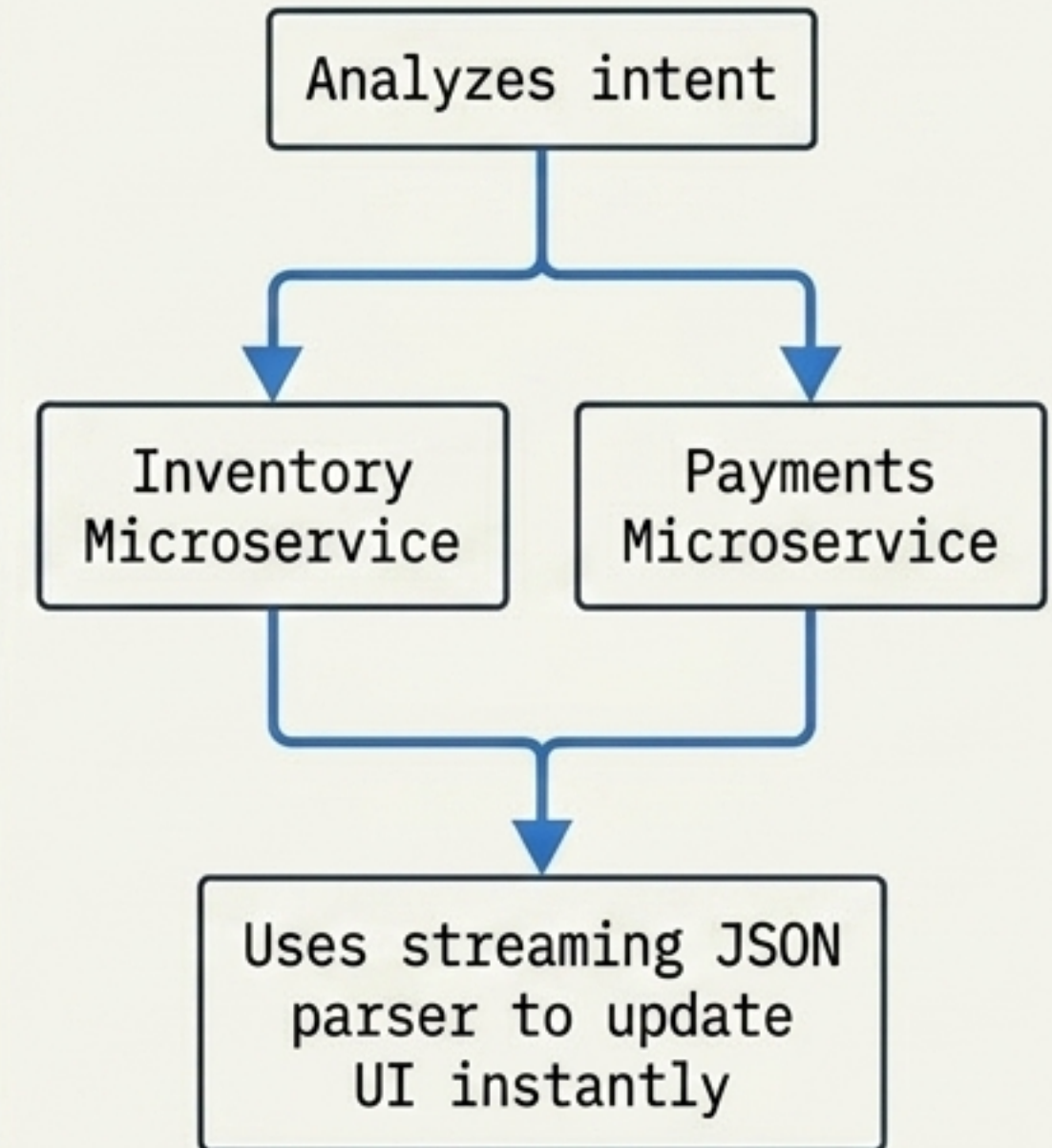
Legal Assistant

Focus: Context Window Engineering



E-Commerce Refund

Focus: Orchestration

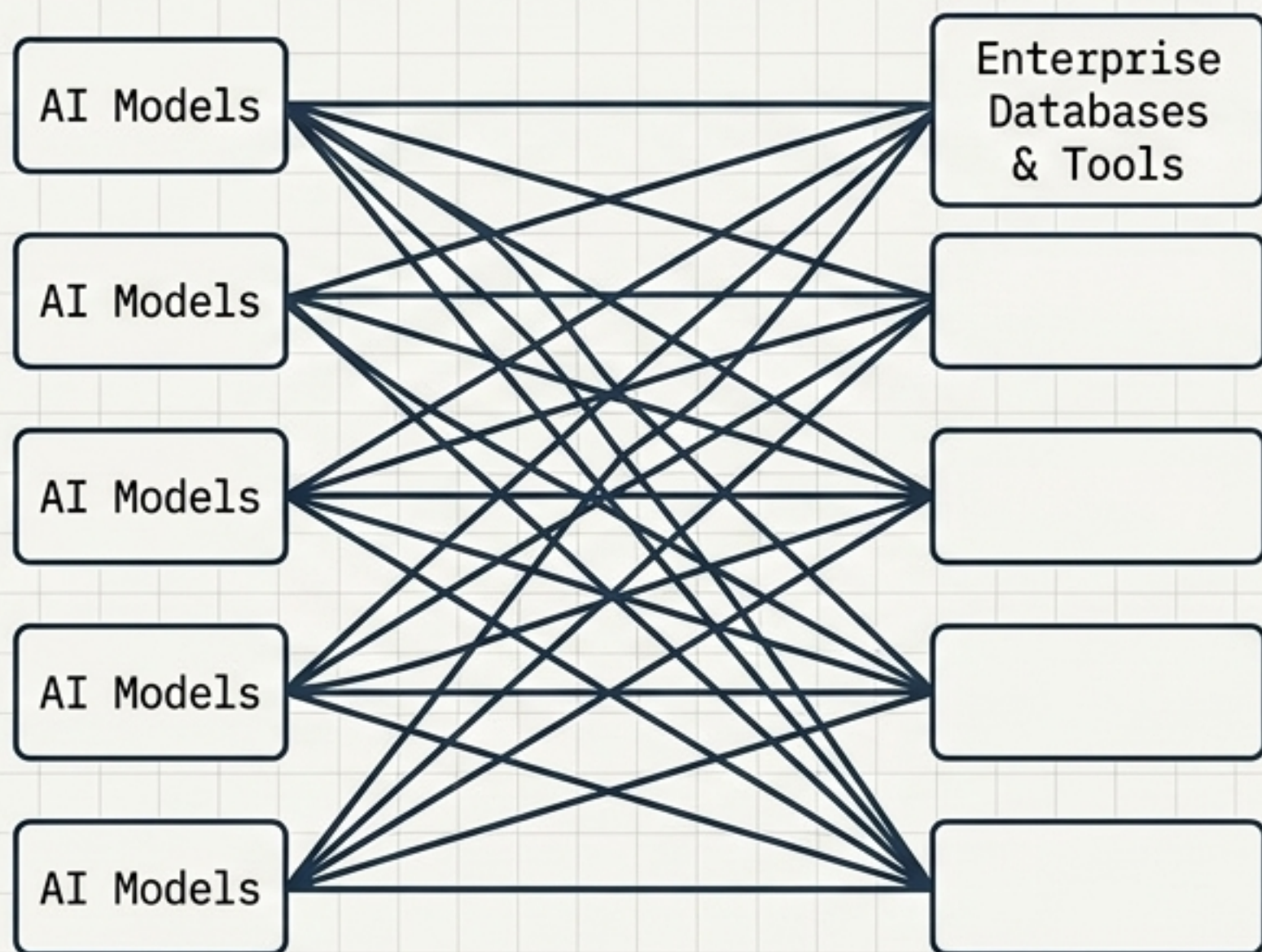


The connectivity evolution matrix

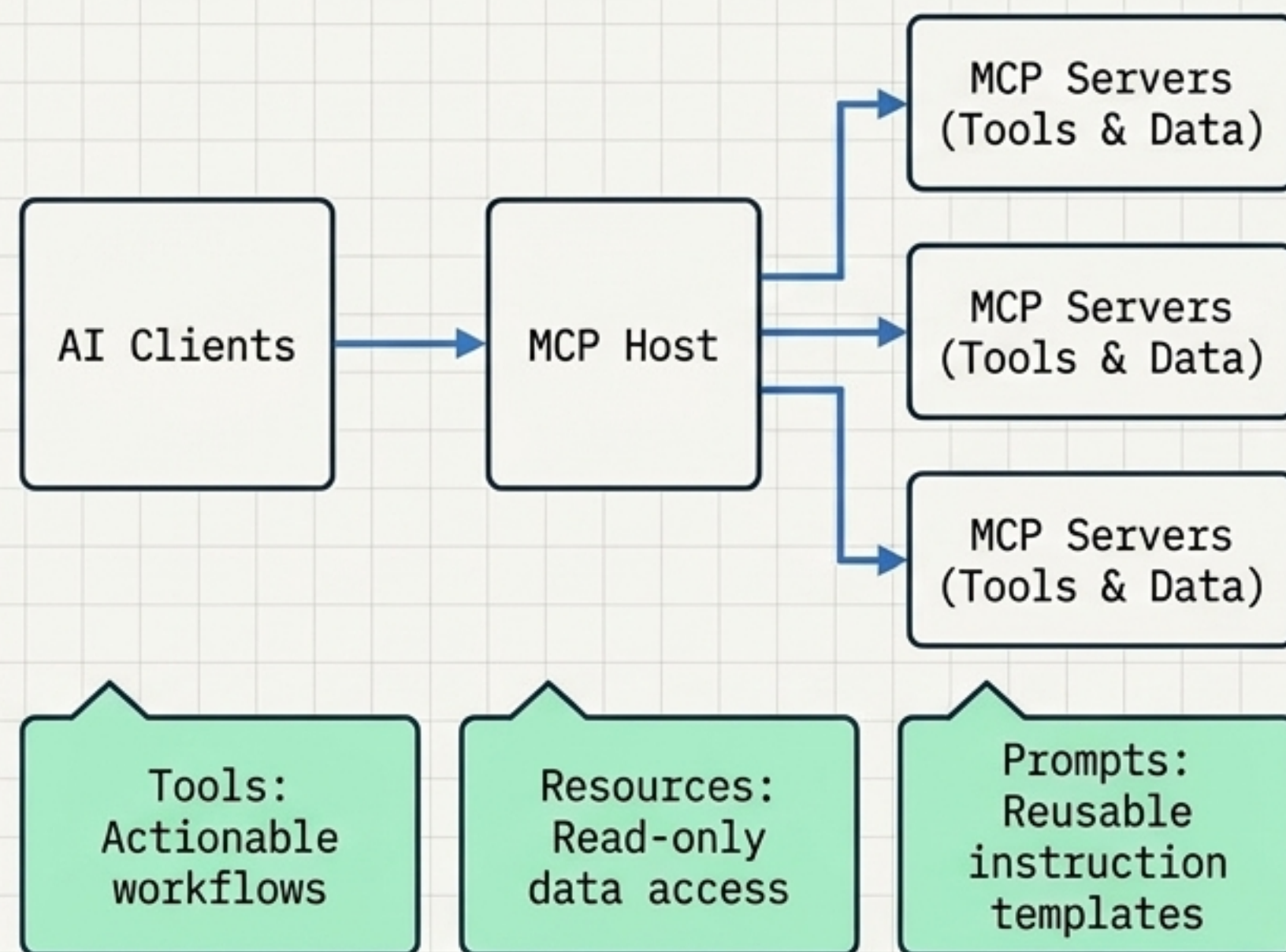
Protocol	Primary Use Case	Format	Risk	AI Suitability
REST	Best for CRUD	JSON	Over-fetching	Low (Requires custom prompt engineering per endpoint)
GraphQL	Best for UI/Client needs	JSON	N+1 query complexity (Requires DataLoaders)	Medium
gRPC	Best for Service-to-Service	Protocol Buffers (Binary)	Strict schema governance required	High (Low network overhead, streaming)
MCP (Model Context Protocol)	Best for AI-to-Enterprise	JSON-RPC 2.0	Emerging standard	Ultimate (Standardized tool/resource discovery)

Eliminating the N x M integration bottleneck with MCP

Before: High engineering effort, fragile coupling



After: JSON-RPC 2.0 Standardization



State persistence, caching, and rate limiting topographies

Redis Caching Strategies

"Cache-Aside"	"Write-Through"	"Write-Behind"
Read-heavy, simple, widely adopted.	Strong consistency, increased write latency.	Very fast user response, risk of data loss on failure.

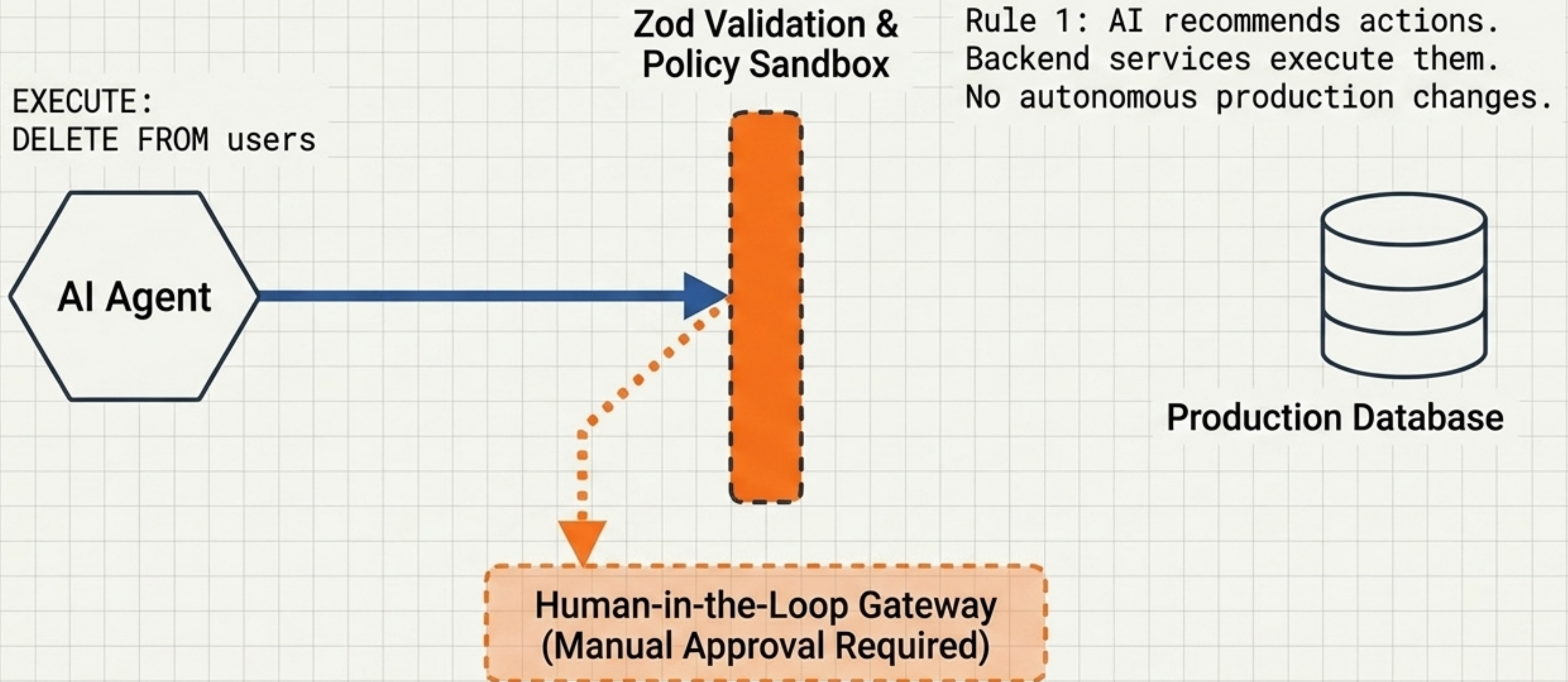
Rate Limiting (Fixed vs. Sliding Window)



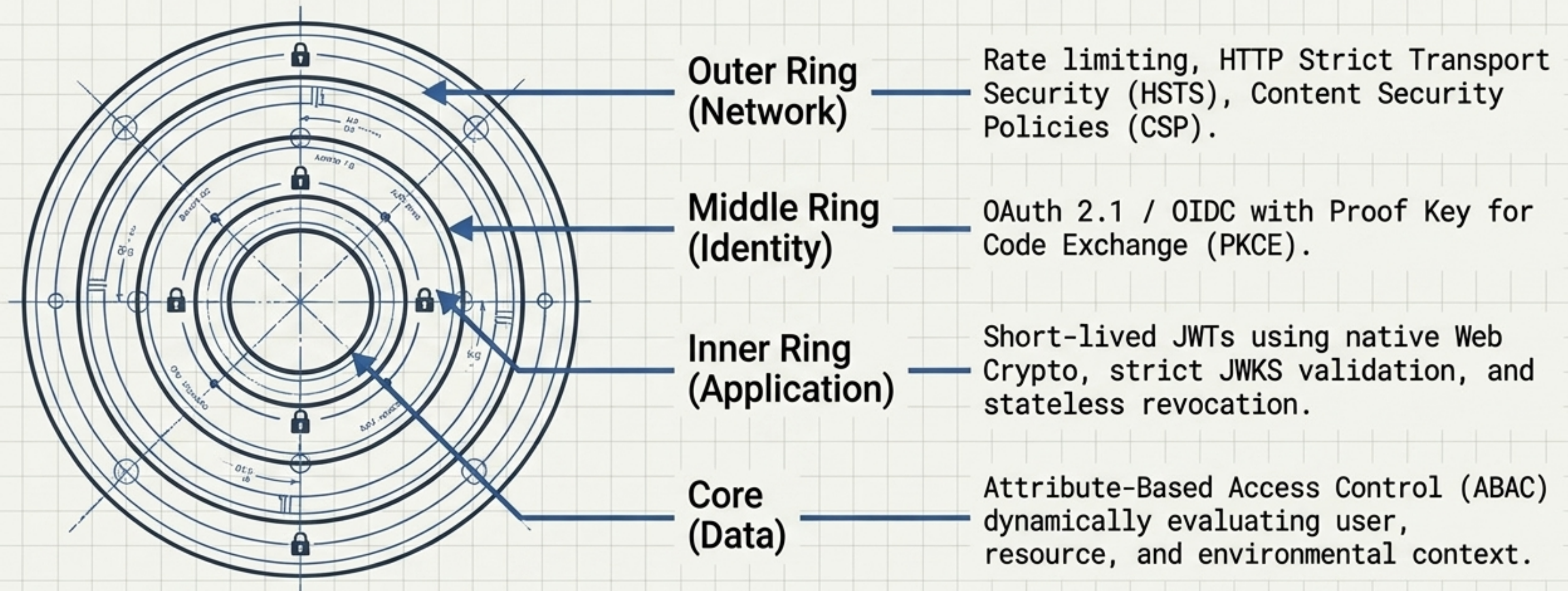
Locking



The guardrail intercepts destructive actions



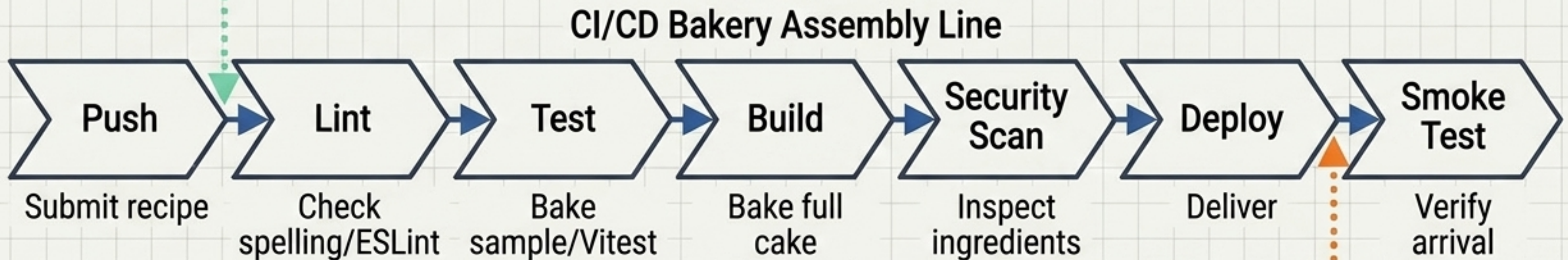
Zero-trust architecture requires layered enforcement



Trust nothing by default. Enforce explicit validation at every boundary transition.

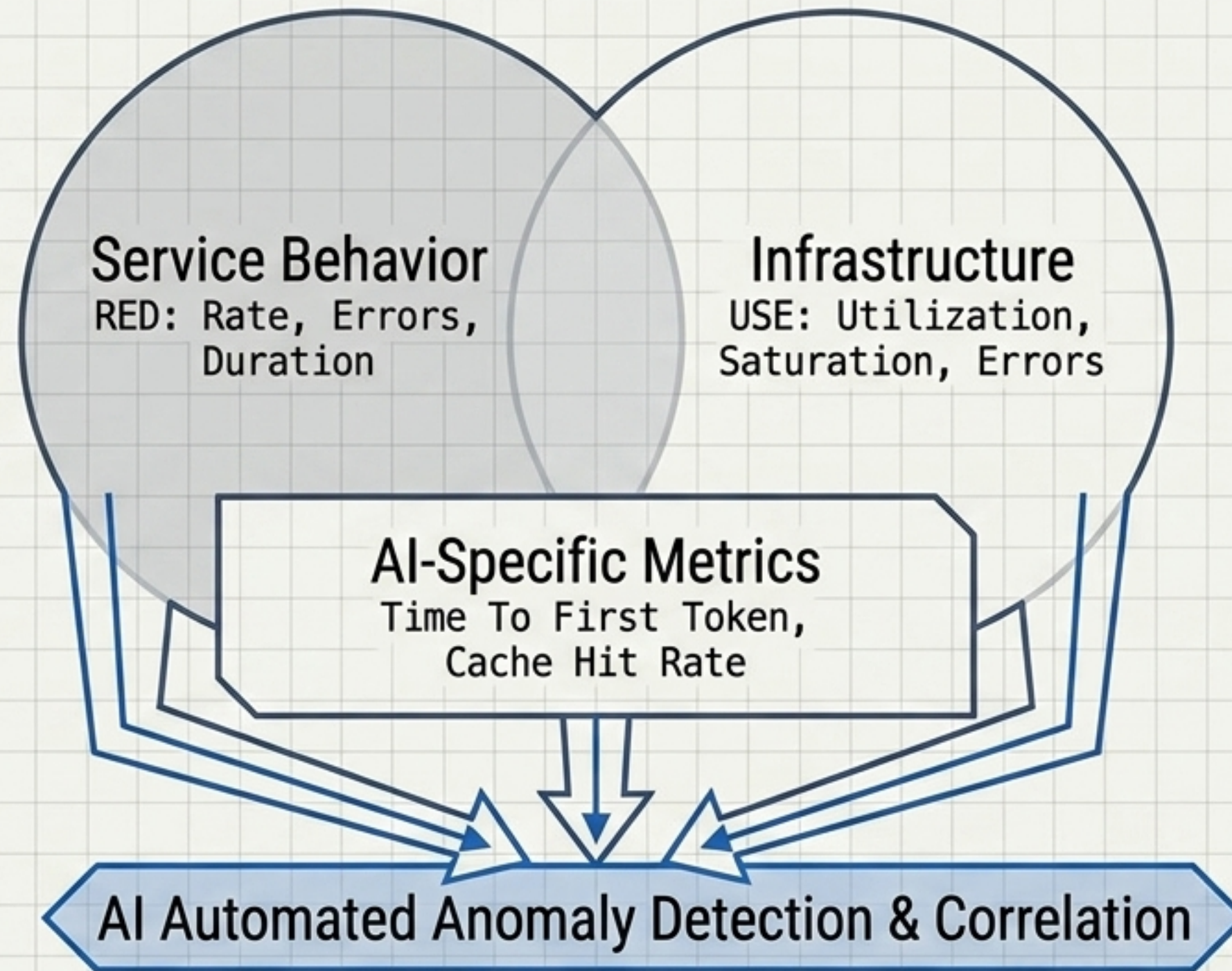
Automated deployment gatekeepers prevent human error

AI Assistance:
Copilot writes the YAML,
humans must review.



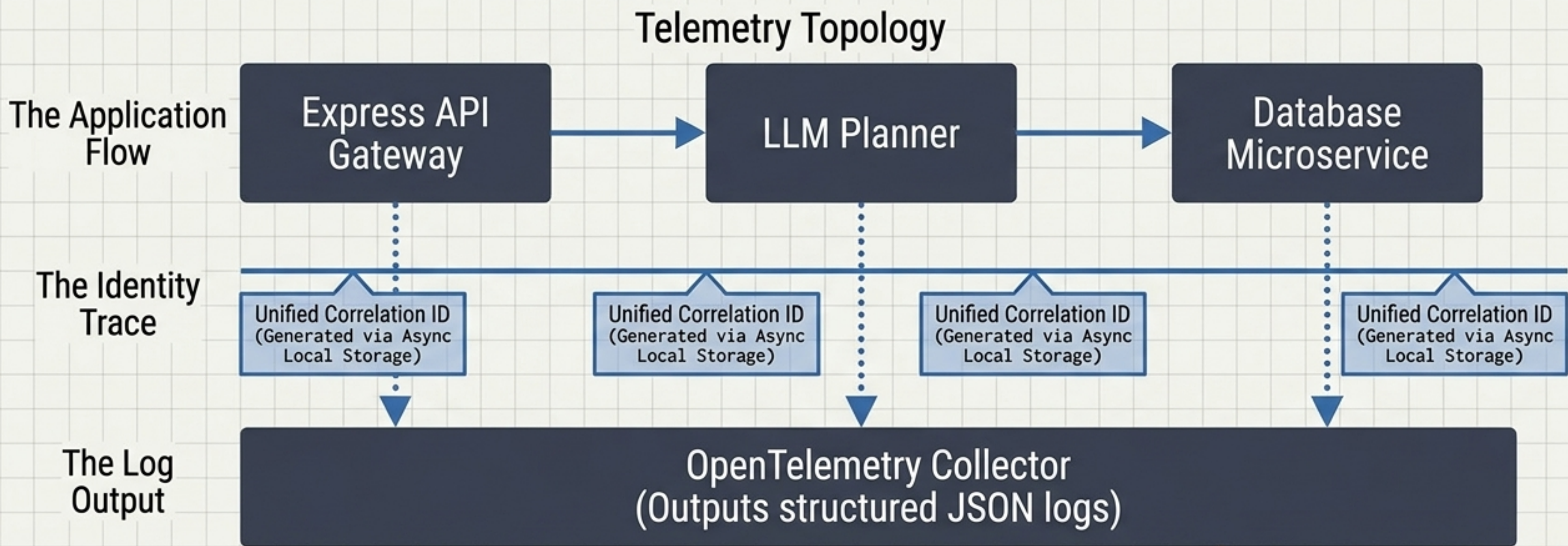
Rollback Strategies: Blue-Green and
Canary deployments ensure MTTM
(Mean Time To Mitigate).

The telemetry triad feeds the AI root-cause engine



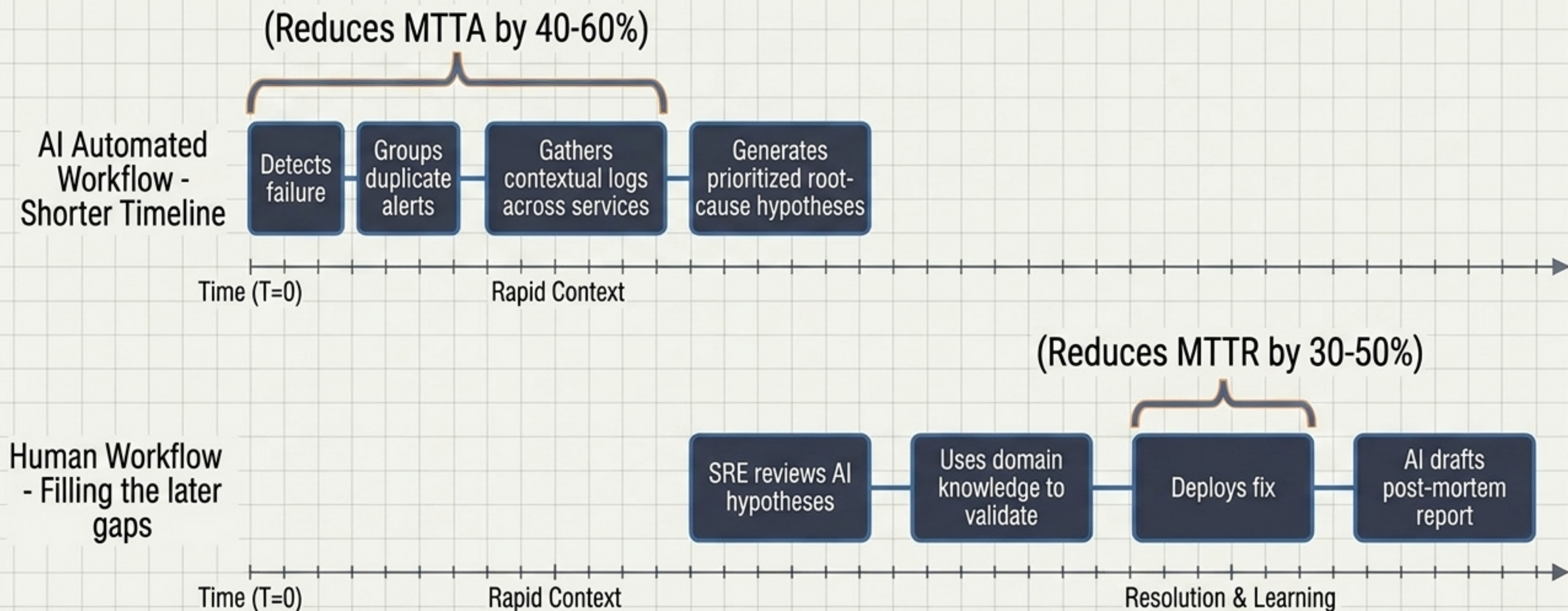
Moving from reactive “grep-and-guess” debugging to dynamic baseline anomaly detection.

OpenTelemetry enables cross-service critical path analysis



Automated PII masking must intercept logs before they reach external AI analysis engines.

Compressing MTTA and MTTR during a production outage



AI handles the context-gathering and hypothesis generation; humans provide the final validation and deployment.

The Full AI-Native Architecture Blueprint

OpenTelemetry Observability Mesh (Feeding logs/traces to AI Diagnostics Engine)

Client Request

Rate-Limited
Gateway
(Redis)

Zod Validation Boundaries

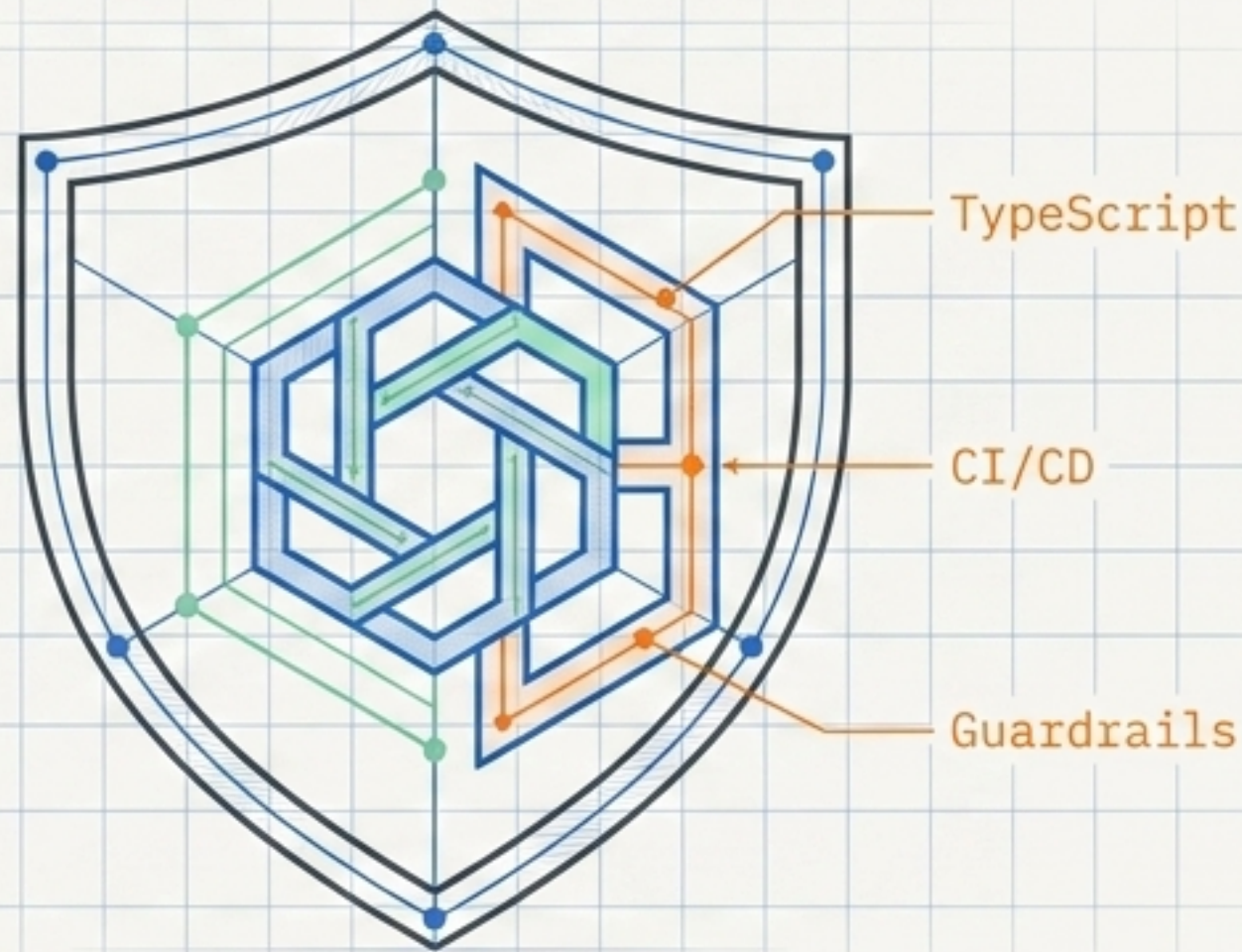
Agent Loop Orchestrator
(Node.js/TypeScript)

MCP Server
(JSON-RPC 2.0)

Enterprise
Data

⚠ Human Approval Queue
(For destructive requests)

Intelligence comes from the model. Reliability comes from the architecture.



**Enterprise AI is an engineering discipline,
not just a prompting exercise.**